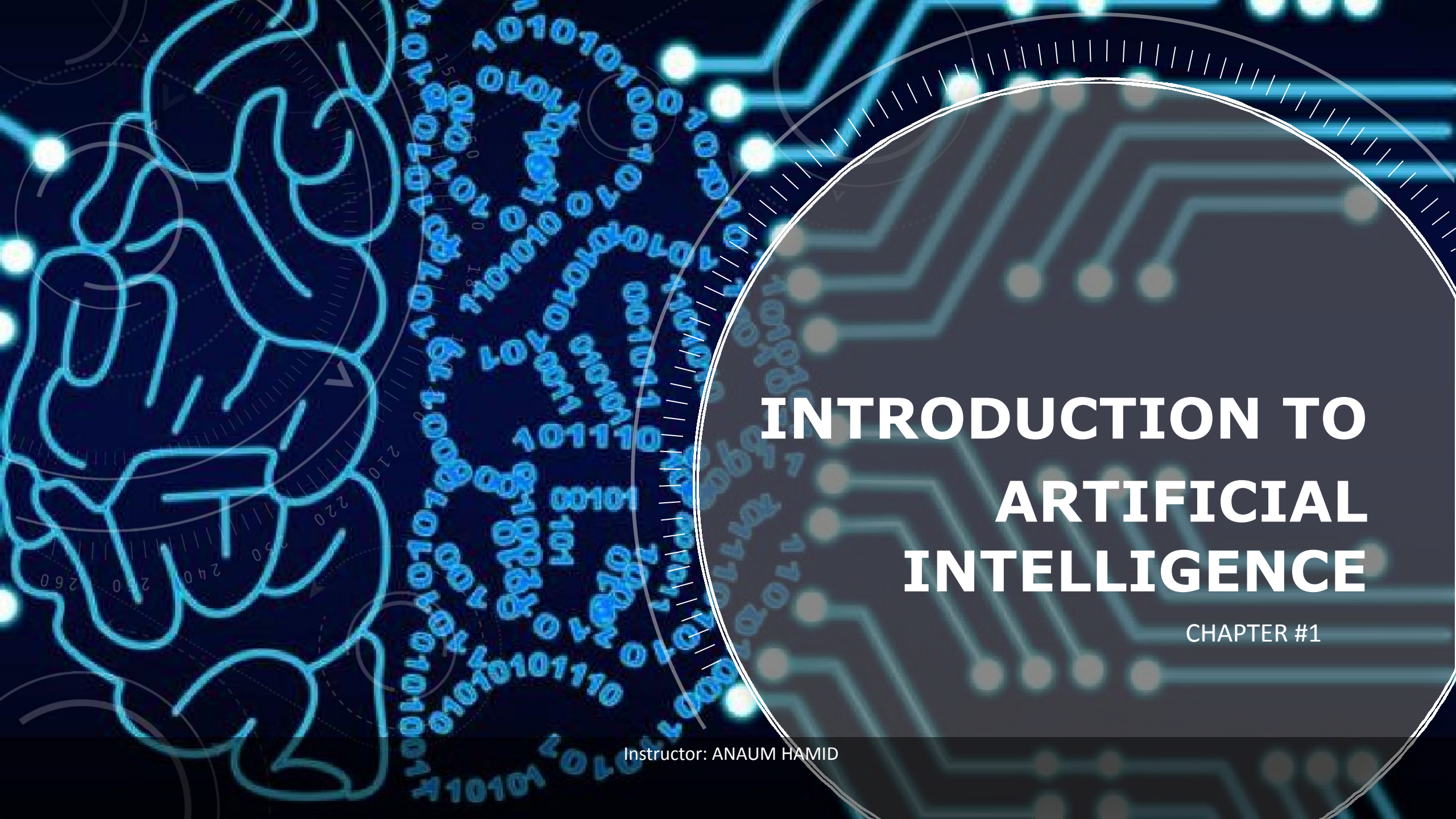




# **INTRODUCTION TO ARTIFICIAL INTELLIGENCE**

CHAPTER #1

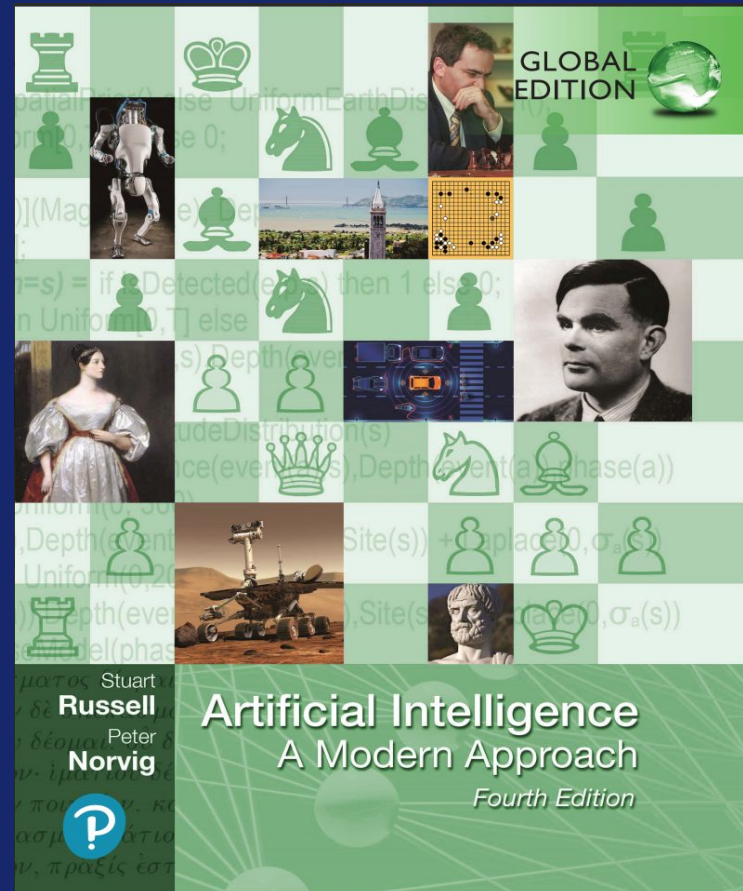
Instructor: ANAUM HAMID

# COURSE CATALOG

1. Introduction and Agents (chapters 1,2)
2. Search (chapters 3,4,5,6)
3. Logic (chapters 7,8,9)
4. Planning (chapters 11,12)
5. Uncertainty (chapters 13,14)
6. Learning (chapters 18,20)
7. Reinforcement Learning



# TEXTBOOK



**Artificial intelligence: A modern approach (4rd edition), Stuart Russell and Peter Norvig**

# MARKS DISTRIBUTION

- **Assignments**            **10 (Best 3 out of 4)**
- **Midterms**                **15 each (x2= 30)**
- **Quiz**                    **10 (Best 3 out of 4)**
- **Final**                    **50**

# AGENDA

1. Introduction to artificial intelligence?
2. A very brief history of AI
3. Foundations of AI
4. AI state of Art
  - How much progress has been made in different aspects of AI

# WHAT MAKES HUMAN INTELLIGENT?

**Learn**

Learning and  
Adapting by  
Experiences from  
Surroundings

**Think**

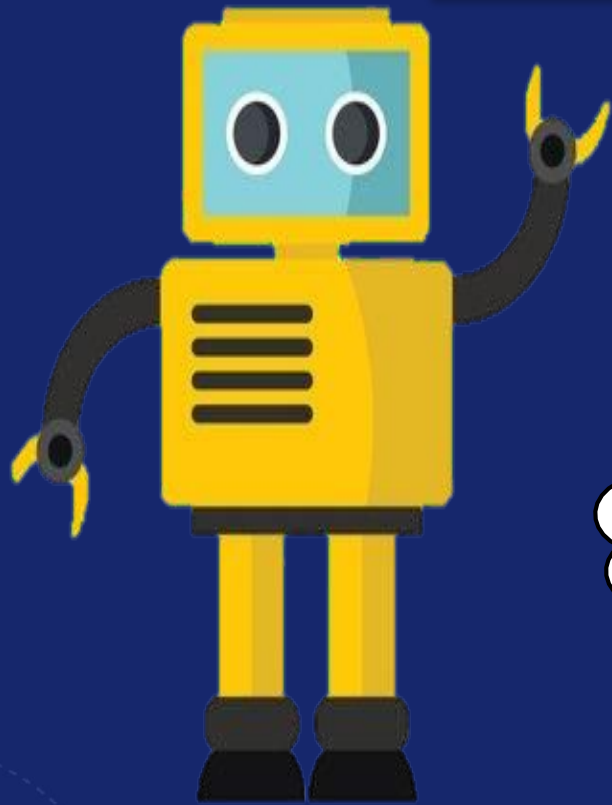
Problem  
solving  
(Game theory)

**Make  
Decisions**

YES/NO



I can show human  
like intelligence



**Think**

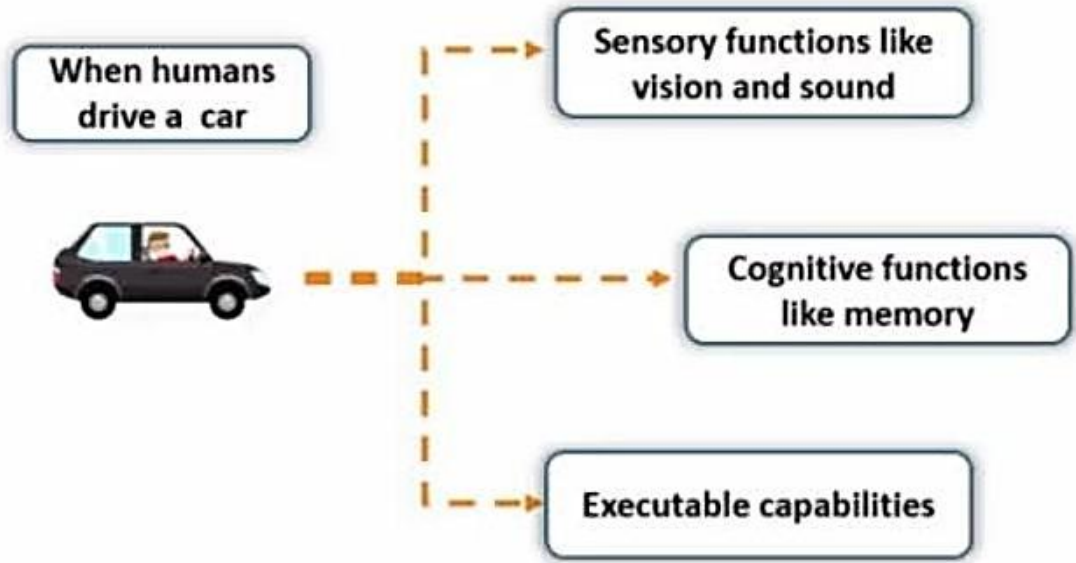
**Make  
Decisions**

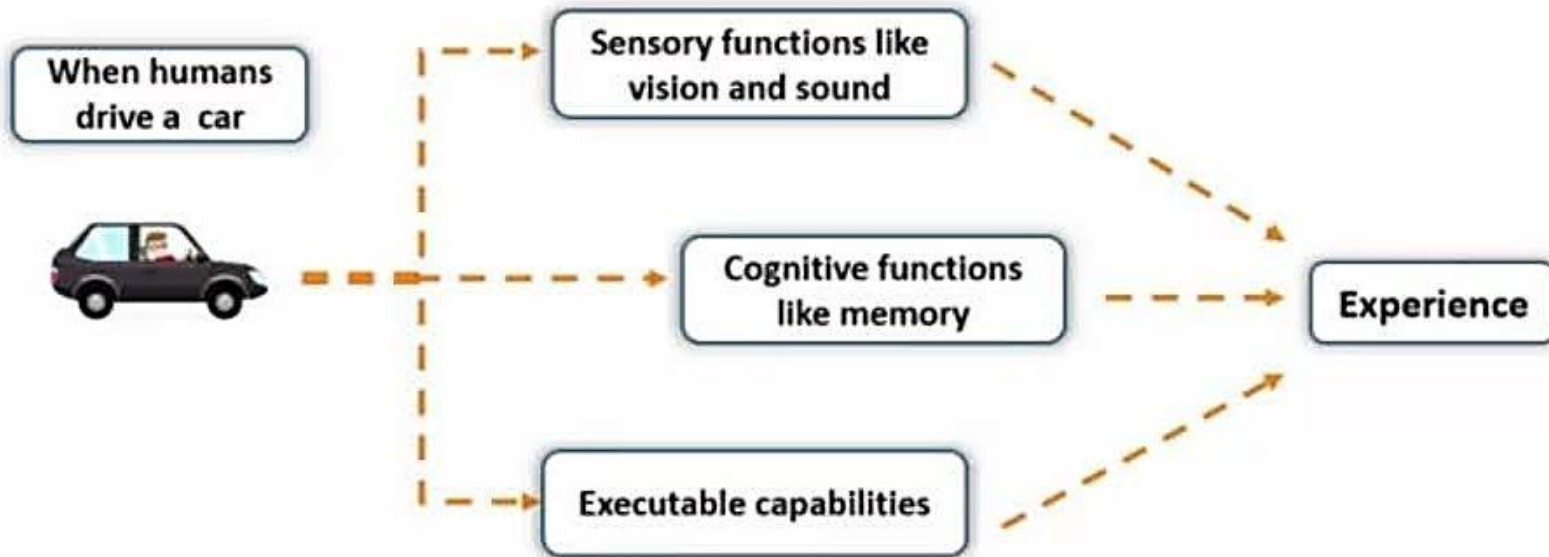
# Human intelligence

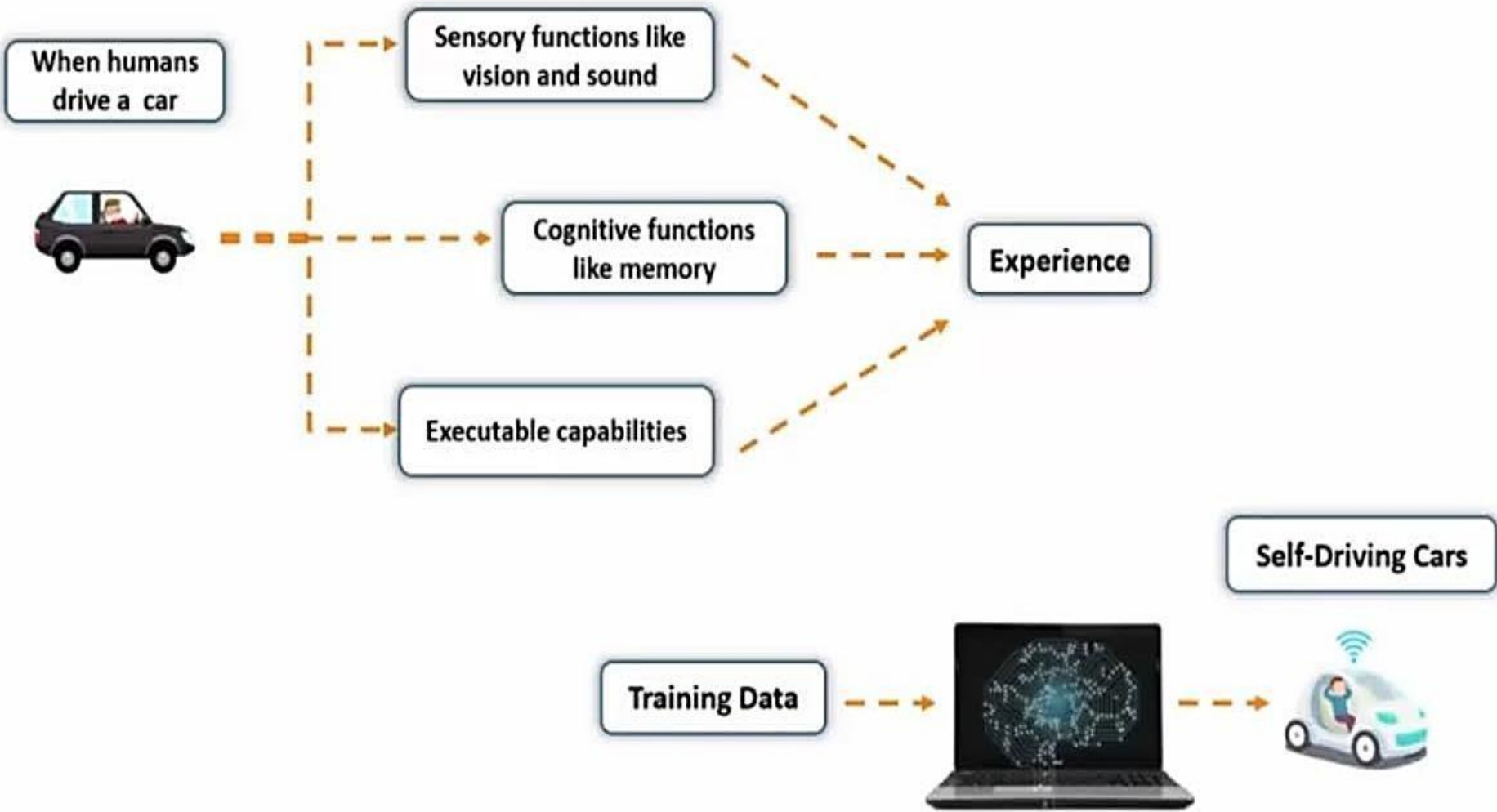
When humans  
drive a car













MOTION FLOW

LANE LINES

LANE LINES

ROAD FLOW

IN-PATH OBJECTS

ROAD LIGHTS

OBJECTS

ROAD SIGNS

RIGHT REARWARD VEHICLE CAMERA





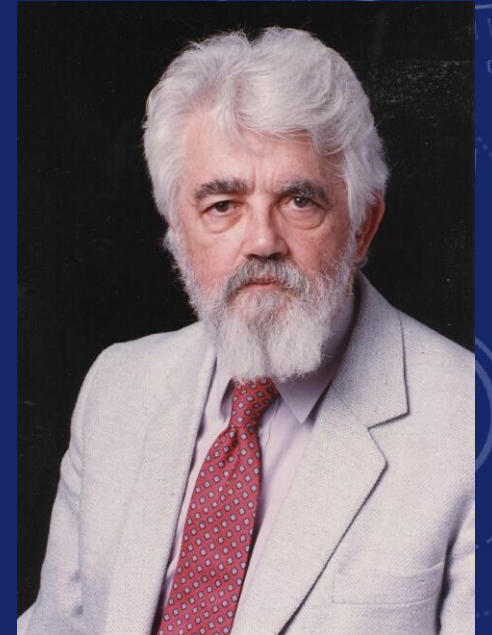
**What is AI ?**

# AI Definition by John McCarthy

(American computer scientist and cognitive scientist)

What is artificial intelligence?

It is the science and engineering of making intelligent machines, especially intelligent computer programs

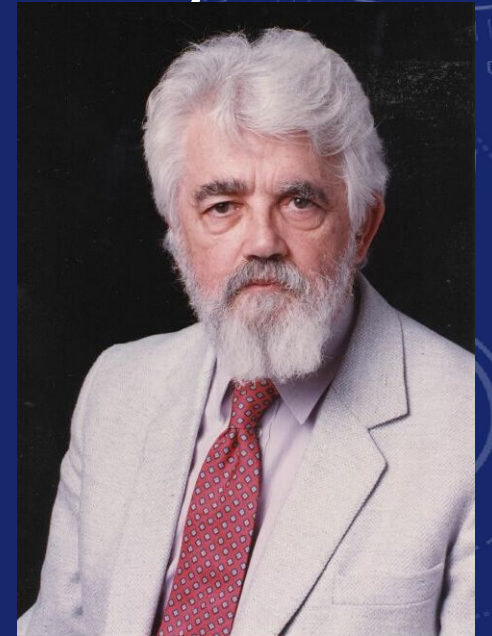


<http://www-formal.stanford.edu/jmc/whatisai/whatisai.html>

# AI Definition by John McCarthy

What is intelligence

Intelligence is the computational part of the ability  
to achieve goals in the world



<http://www-formal.stanford.edu/jmc/whatisai/whatisai.html>

## What is an AI?

“The study and design of intelligent agents, where an intelligent agent is a system that perceives its environment and takes actions that maximize its chances of success.”

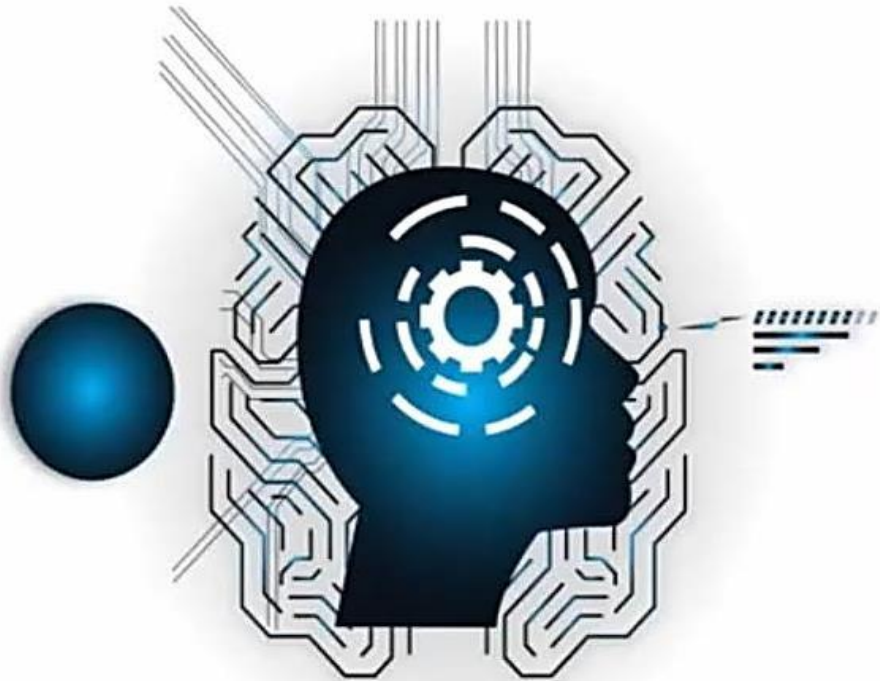
Russel and Norvig, AIMA  
book



# HISTORY OF AI

- 1950: Turing
  - Turing's "Computing Machinery and Intelligence"
- 1956: birth of AI
  - Dartmouth meeting: "Artificial Intelligence" name adopted
- 1950s: initial promise
  - Early AI programs, including
  - Samuel's checkers program
  - Newell & Simon's Logic Theory
- 1986-- Rise of machine learning
  - Neural networks return to popularity
  - Major advances in machine learning algorithms and applications
- 1990-- Role of uncertainty
  - Bayesian networks as a knowledge representation framework
- 1995-- AI as Science
  - Integration of learning, reasoning, knowledge representation
  - AI methods used in vision, language, data mining, etc

# Artificial Intelligence



**Intelligence displayed by machines**

**OR**

**The capacity of a machine to imitate  
intelligent human behaviour**

# ML and DL



A bunch of statistical algorithms and tools to learn from the data

Connected with algorithms inspired by the structure and function of the brain called "Artificial Neural Networks"

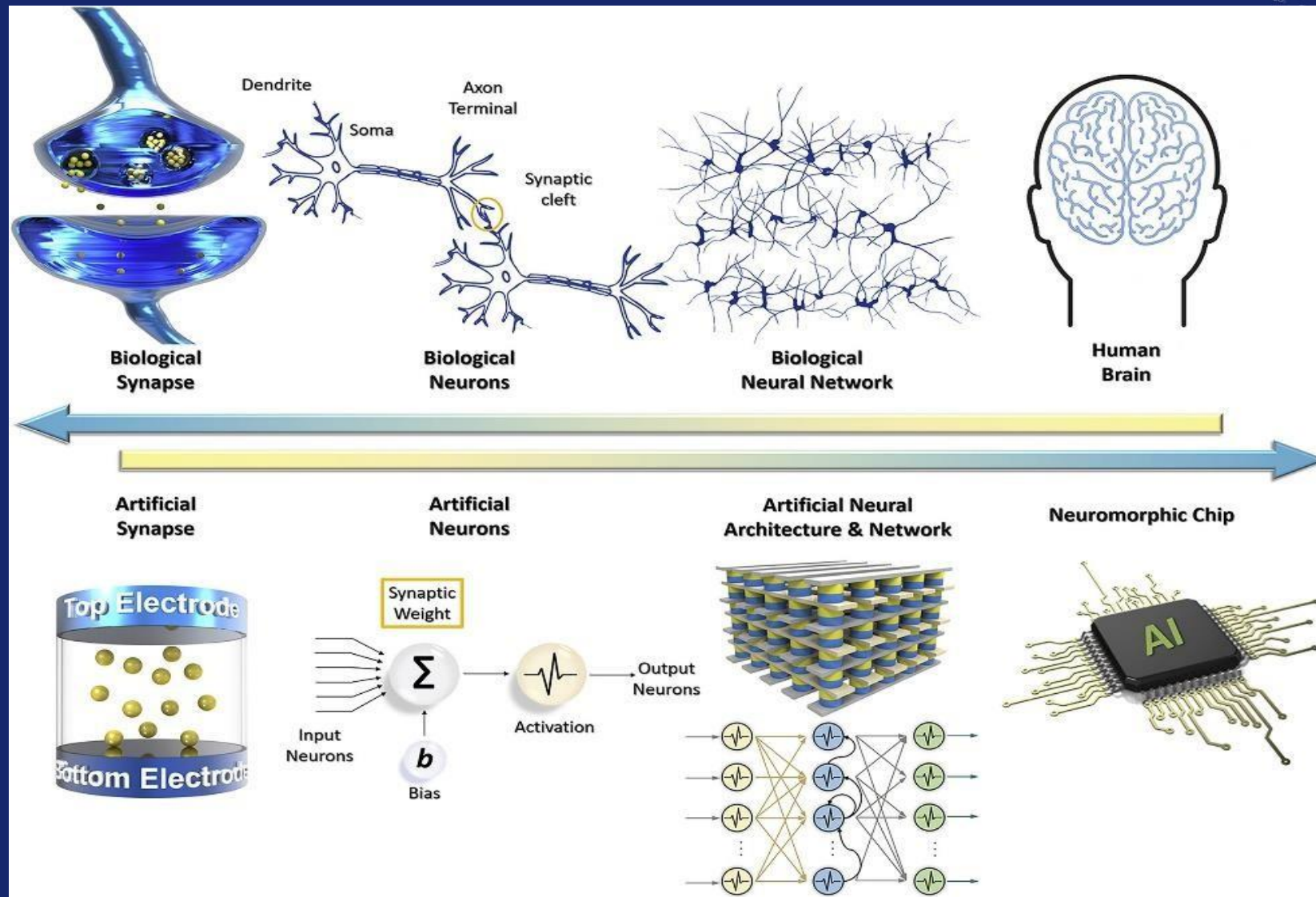


# Neuroscience Cell Components

A **neuron** is a specialized nerve cell that receives, processes, and transmits information through electrical and chemical signals in the nervous system.

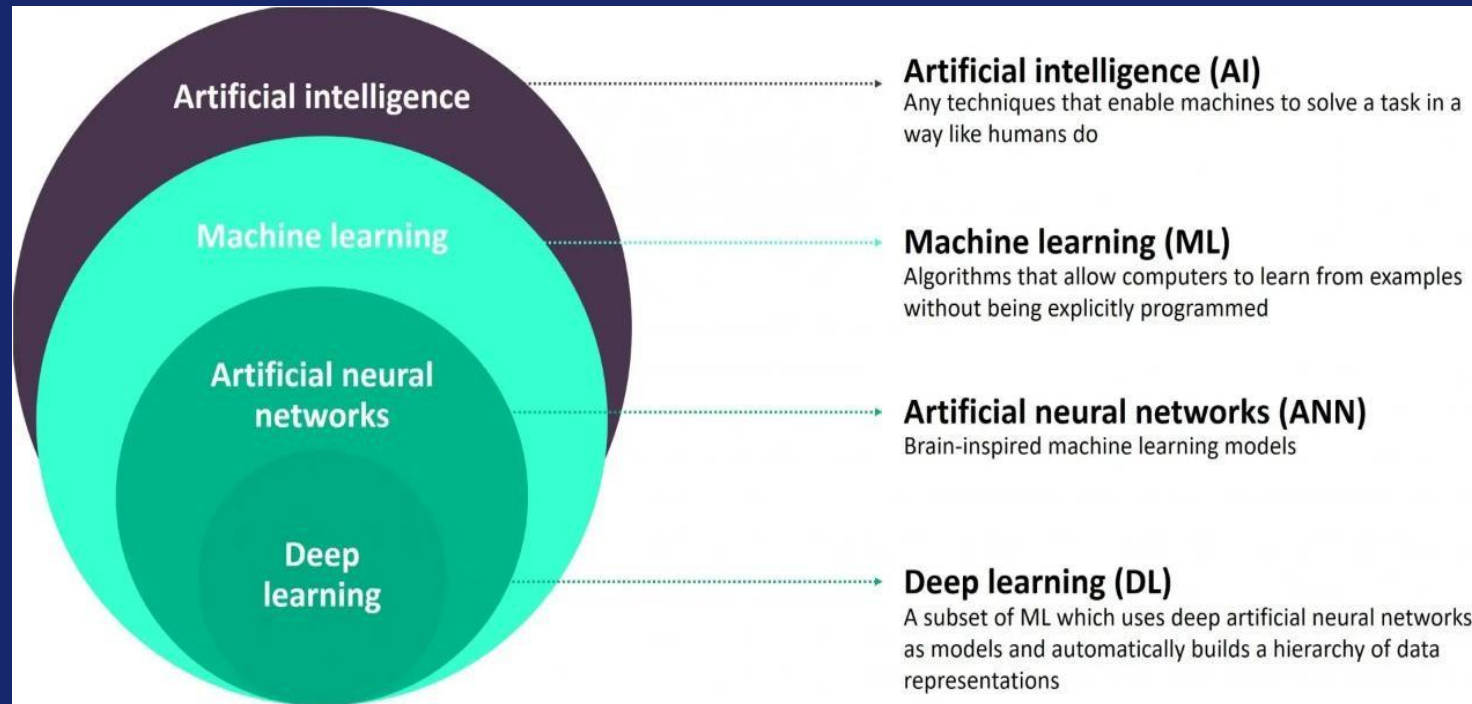
- Dendrites: Receive incoming signals from other neurons.
- Soma (cell body): Integrates signals and maintains the neuron's functions.
- Axon: Transmits electrical impulses away from the soma to other neurons.

# DL (BIOLOGICAL NEURON)



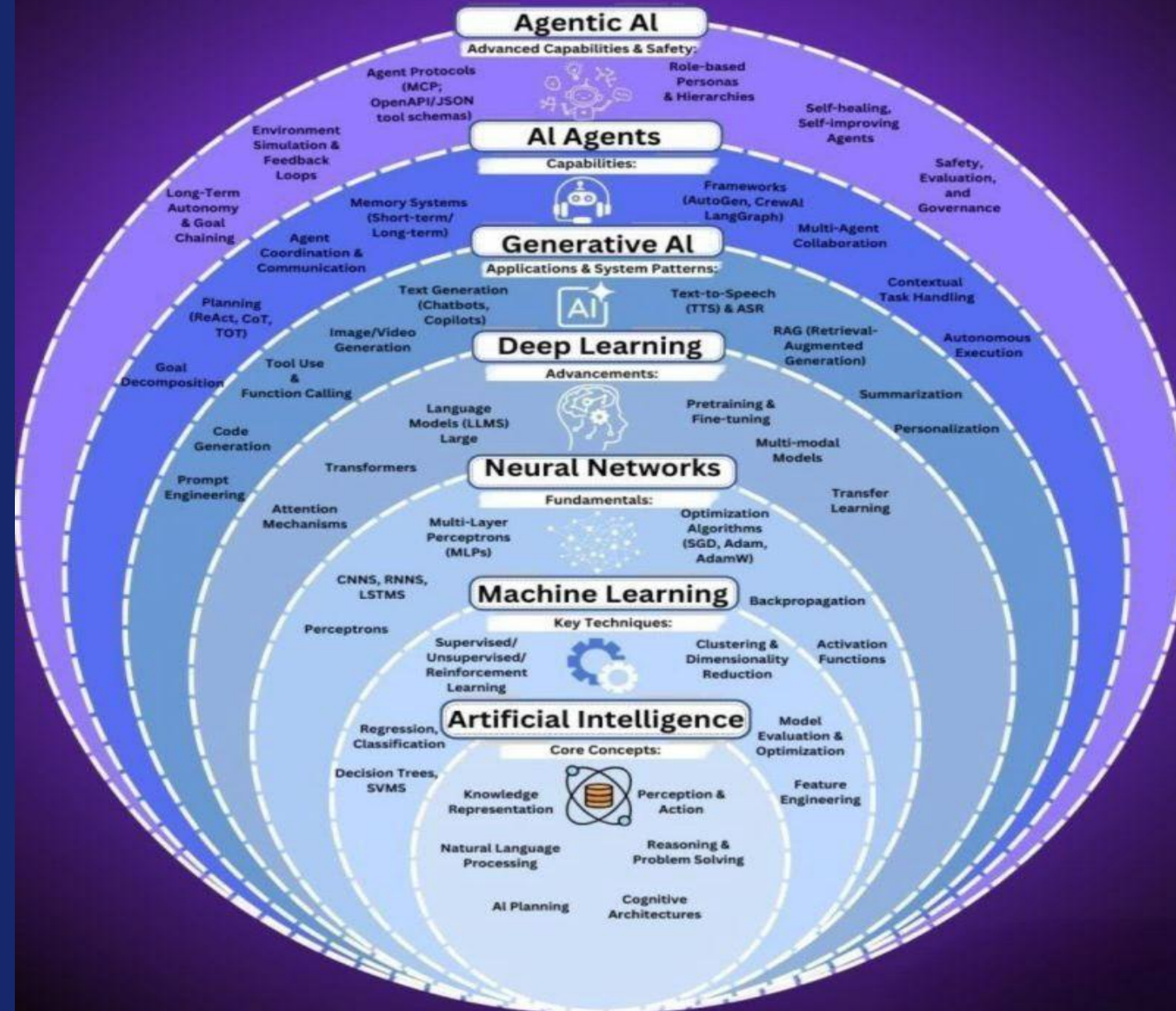


# (BIOLOGICAL NEURON)



| Biological Neuron  | Artificial Neuron |
|--------------------|-------------------|
| Dendrites          | Input             |
| Cell Nucleus(Soma) | Node              |
| Axon               | Output            |
| Synapse            | Interconnections  |

# The Agentic AI Universe



# ARTIFICIAL NEURON

Training set

Inputs Output

000 001

001 010

010 011

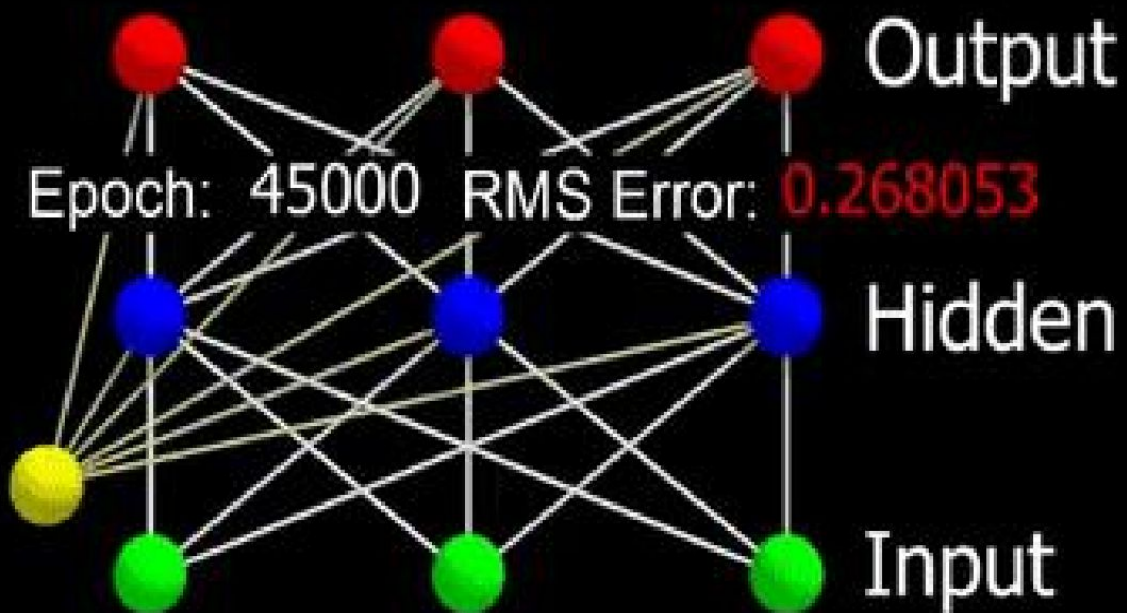
011 100

100 101

101 110

110 111

111 000





### Train Data



Feature vector

(170, 35, 169, 51, 38, ...)

Label

(Bicycle)

(86, 79, 50, 181, 25, ...)

(Boat)

(13, 157, 90, 178, 145, ...)

(Car)

(94, 90, 202, 25, 158, ...)

(Plane)

### Test Data



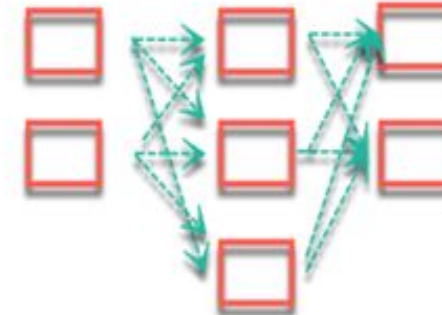
Feature vector

(213, 167, 7, 54, 23, ...)

Label

(?)

Deep learning  
3 connected layers



Inference



Output

(Car)

# STATE OF THE ART

- Deep Blue defeated the reigning world chess champion Garry Kasparov in 1997
- AI (EQP) program proved a mathematical conjecture (Robbins conjecture) unsolved for Decades in 1996 developed by William McCune
- NASA's on-board autonomous planning program, like CASPER, controlled the scheduling of operations for a spacecraft
- Speech recognition: A traveler calling United Airlines to book a flight can have the entire conversation guided by an automated speech recognition and dialog management system.
- Face recognition software available in consumer cameras



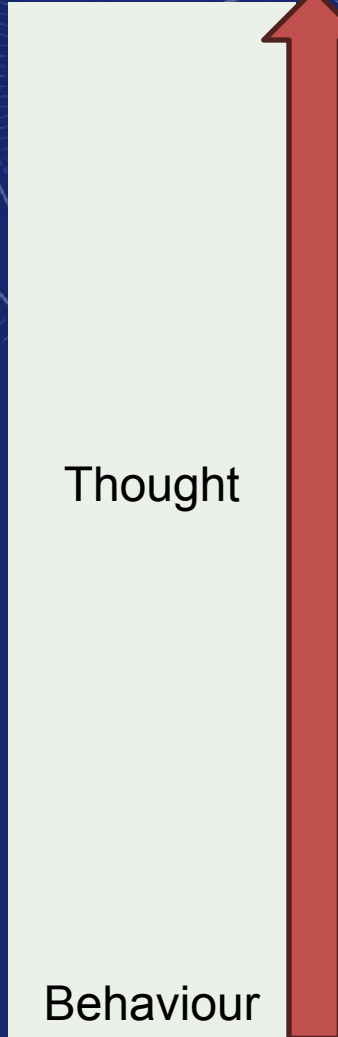
# WHAT IS ARTIFICIAL INTELLIGENCE?

It is the science and engineering of making intelligent machines, especially intelligent computer programs. It is related to the similar task of using computers to understand human intelligence, but AI does not have to confine itself to methods that are biologically observable.

Views of AI fall into four categories:

- Thinking Humanly
- Acting Humanly
- Thinking Rationally
- Acting Rationally

# Definitions



Human performance metric: involving observations and hypothesis

Ideal or rational performance metric: combination of mathematics and engineering



# What is AI?

## Four schools of thoughts (Russel & Norvig)

| Thinking humanly                                                                                                                      | Thinking rationally                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <p>“The exciting new effort to make computers think... machines with minds, in the full and literal sense.”<br/>(Haugeland, 1985)</p> | <p>“The study of mental faculties through the use of computational models.”<br/>(Charniak &amp; McDermott, 1985)</p> |
| Acting humanly                                                                                                                        | Acting rationally                                                                                                    |
| <p>“The study of how to make computers do things which, at the moment, people are better.”<br/>(Rich &amp; Knight, 1991)</p>          | <p>“Computational Intelligence is the study of the design of intelligent agents.”<br/>(Poole et al., 1998)</p>       |

# Four Approaches to AI

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## Acting Humanly

The Turing Test approach. Can a machine communicate indistinguishably from a human?



## Thinking Rationally

The "Laws of Thought" approach. Using formal logic (syllogisms) to arrive at correct conclusions.



## Thinking Humanly

The Cognitive Modeling approach. Creating systems that think like humans based on psychological evidence.



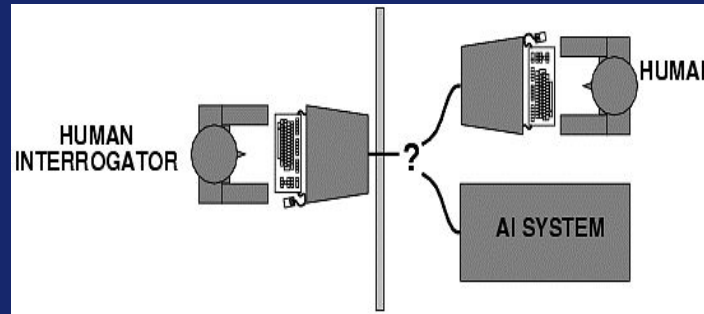
## Acting Rationally

The **Rational Agent** approach. Achieving the best outcome. This is the focus of the book.



# ACTING HUMANLY: TURING TEST

- Alan Turing (1950) "Computing machinery and intelligence"
- "Can machines think?" ⇔ "Can machines behave intelligently?"
- Operational test for intelligent behavior: the Imitation Game



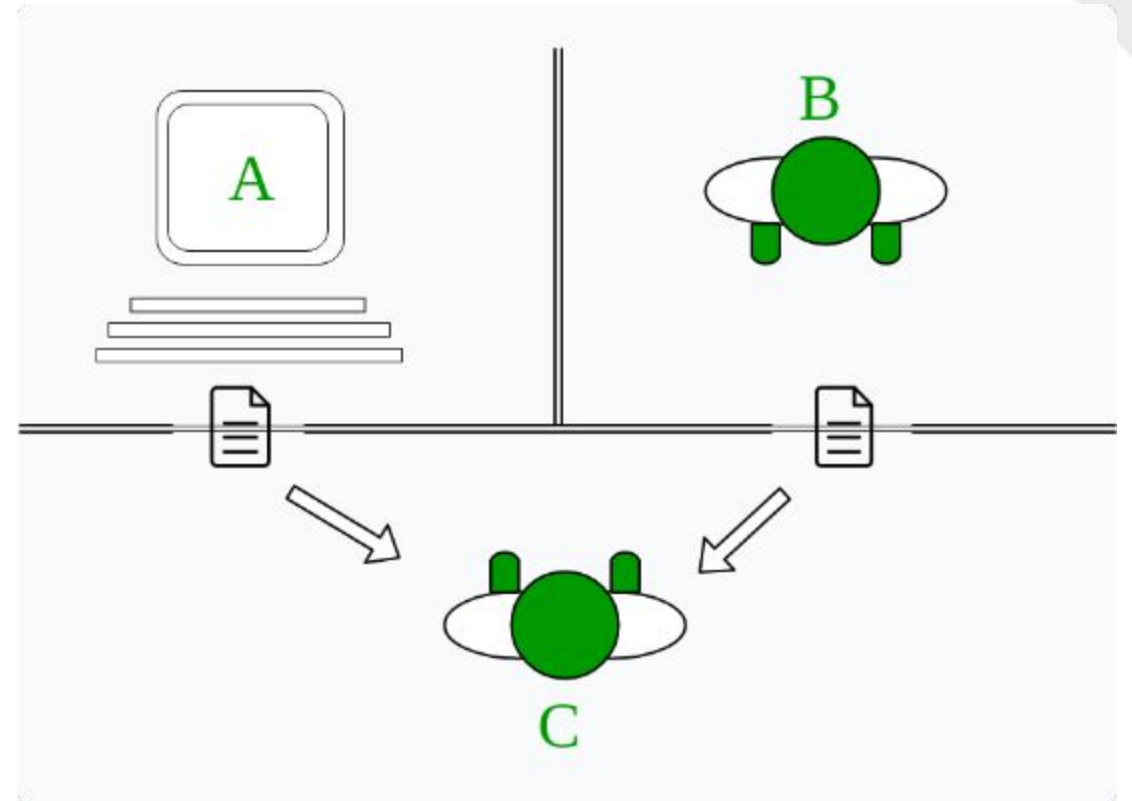
It was designed to provide a satisfactory operational definition of intelligence. A computer passes the test if a human interrogator, after posing some written questions, cannot tell whether the written responses come from a person or from a computer.

# Acting Humanly: The Turing Test

## Proposed by Alan Turing (1950)

The Turing Test was designed to provide a satisfactory operational definition of intelligence.

- ✓ A computer passes the test if a human interrogator cannot distinguish it from a human being.
- ✓ Requires capabilities in:
  - ✓ Natural Language Processing
  - ✓ Knowledge Representation
  - ✓ Automated Reasoning
  - ✓ Machine Learning



# THINKING HUMANLY: THE COGNITIVE MODELING APPROACH

- Requires detailed **matching of computer behavior and timing** to detailed **measurements of human subjects** gathered in psychological experiments.
- Cognitive Science approach
  - Try to get “inside” our minds
  - E.g., conduct experiments with people to try to “reverse-engineer” how we reason, learning, remember, predict
- So, we need to get **inside the actual workings of human minds**.
- There are three ways to do this:
- through **introspection**—trying to catch our own thoughts as they go by;
- through **psychological experiments**—observing a person in action;
- and through **brain imaging**—observing the brain in action.

# Thinking Humanly: Cognitive Modeling

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## The "Inside" Approach

To say a program thinks like a human, we must know how humans think. This requires scientific theories of internal activities of the brain.

Validation methods:

1. **Introspection:** Catching our own thoughts.
2. **Psychological Experiments:** Observing a person in action.
3. **Brain Imaging:** Observing the brain in action.

## Cognitive Science

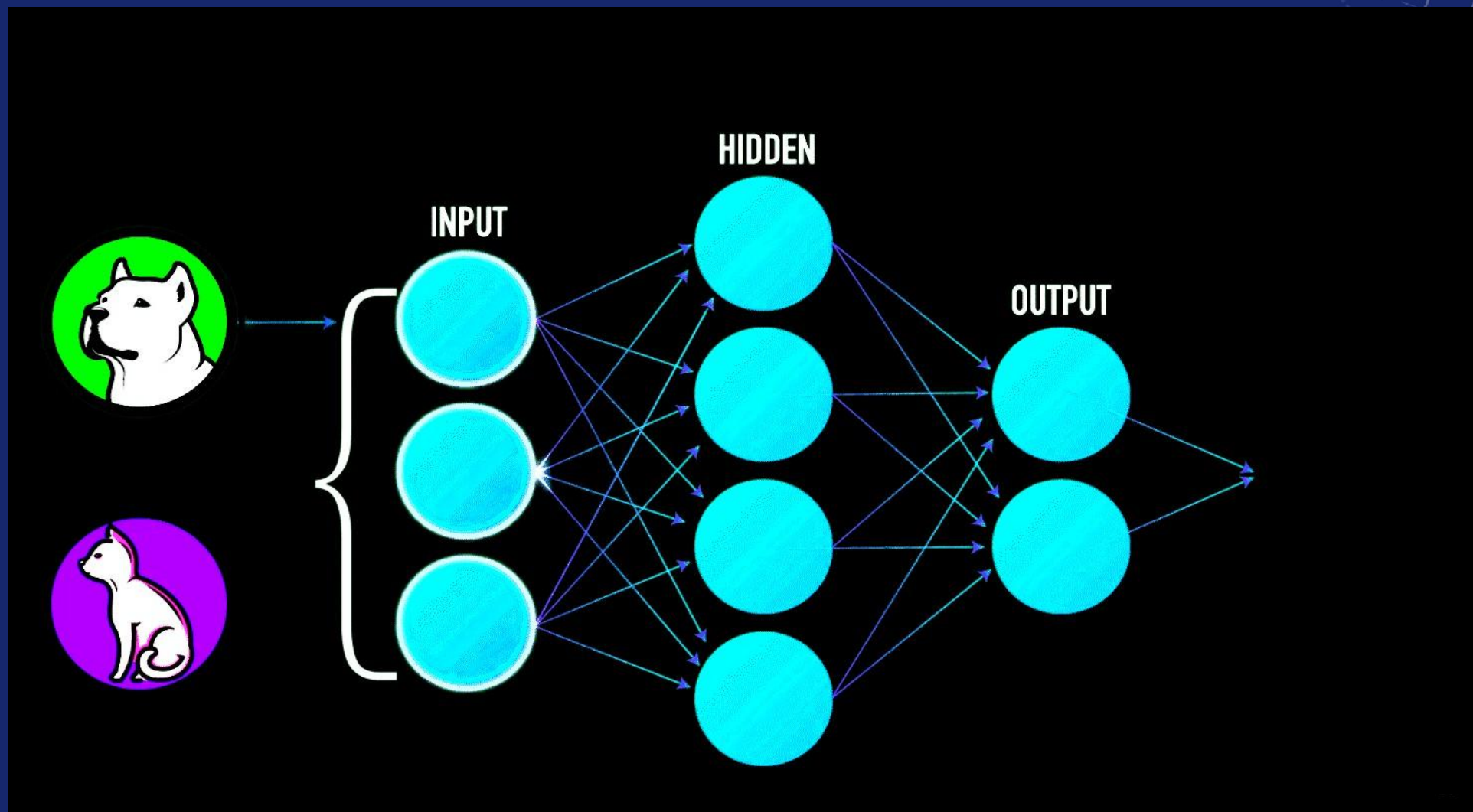
This interdisciplinary field brings together computer models from AI and experimental techniques from psychology to construct precise and testable theories of the human mind.

the pigeon experiment classic box-and-banana problem.

<https://www.youtube.com/watch?v=mDntbGRPeEU>







# THINKING RATIONALLY: LAWS OF THOUGHT

- Formalize “correct” reasoning using a mathematical model (e.g. of deductive reasoning/logical conclusion).
- Logics Program: Encode knowledge in formal logical statements and use mathematical deduction to perform reasoning:
- **Problems:**
  - -Formalizing common sense knowledge is difficult.
  - -General deductive inference is computationally intractable.

# Thinking Rationally: Laws of Thought

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*"Socrates is a man;*

*All men are mortal;*

*Therefore, Socrates is mortal."*

— Aristotle (Syllogism)

The "laws of thought" approach relies on **Logic**. The goal is to codify "right thinking" into irrefutable logical processes. However, it is difficult to take informal knowledge and state it in formal terms required by logical notation.

# ACTING RATIONALLY: THE RATIONAL AGENT

- An agent is an entity that perceives its environment and is able to execute actions to change it.
- Agents have inherent goals that they want to achieve (e.g. survive, reproduce).
- A **rational agent** acts in a way to **maximize the achievement of its goals**.
- True maximization of goals requires omniscience and unlimited computational abilities.
- Limited rationality involves maximizing goals within the computational and other resources available.

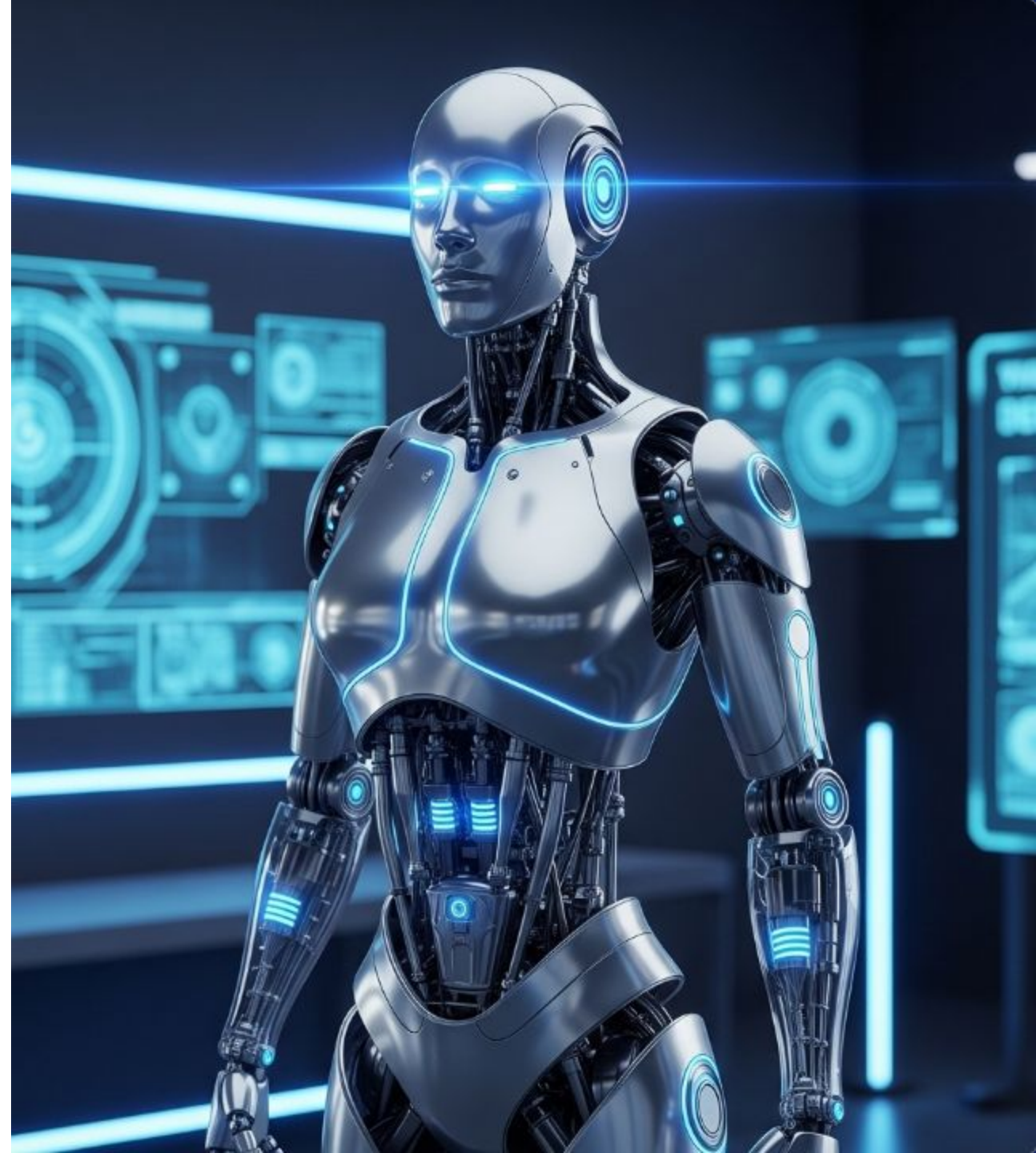
# Acting Rationally: The Rational Agent

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This is the core approach of the book.

An *agent* is just something that acts. But a computer agent is expected to do more: operate autonomously, perceive its environment, persist over a prolonged time period, adapt to change, and create and pursue goals.

**Definition:** A rational agent is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.

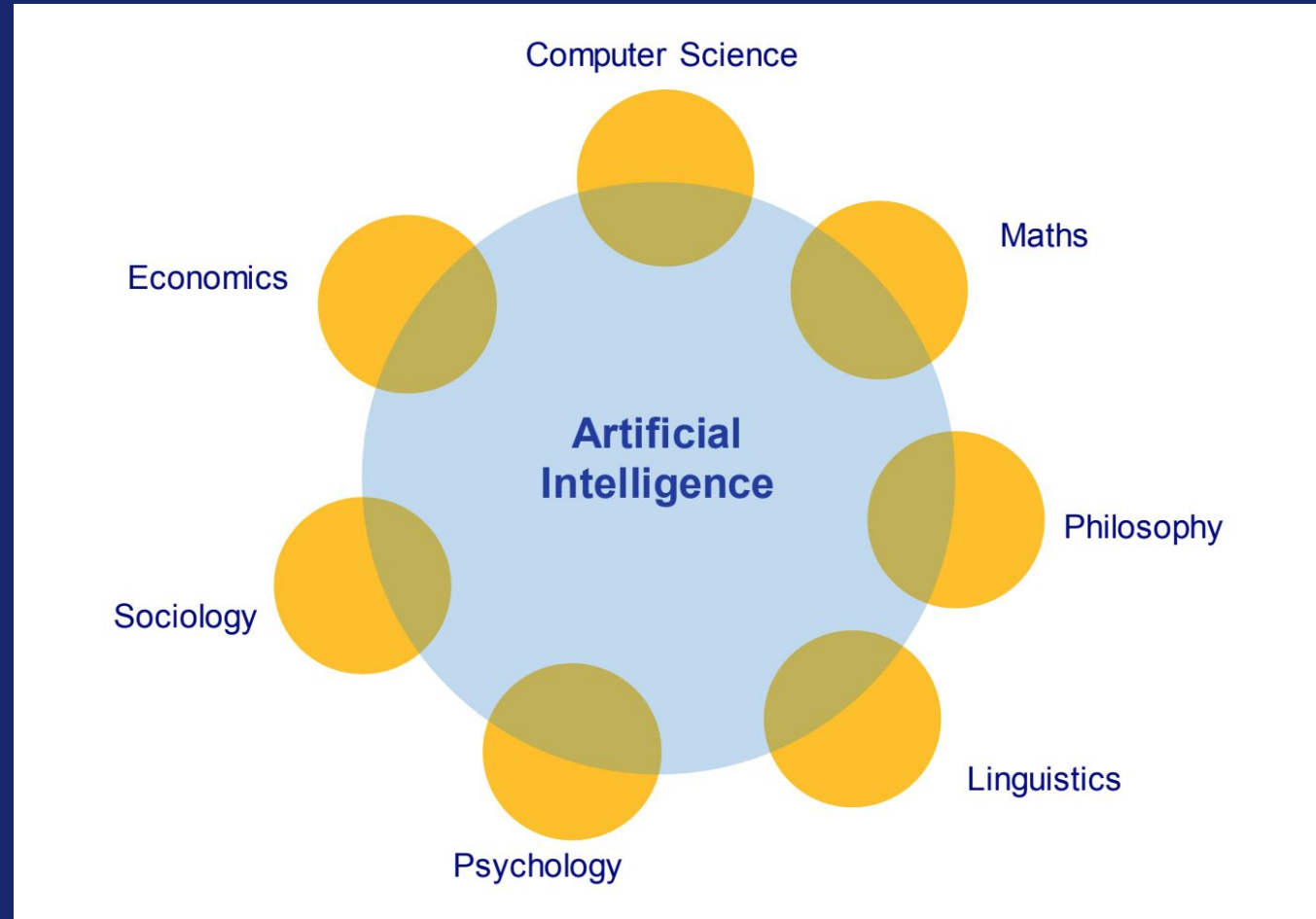




# FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

- **Philosophy** Logic, methods of reasoning, mind as physical system, foundations of learning, language , rationality.
- **Mathematics** formal representation and proof, algorithms , computation.
- **Probability/statistics** modeling uncertainty, learning from data
- **Economics** utility, decision theory, rational economic agents
- **Neuroscience** neurons as information processing units.
- **Computer engineering** building fast computers
- **Psychology** how do people behave, perceive, process cognitive science information, represent knowledge.
- **Linguistics** knowledge representation, grammars

# FOUNDATIONS OF ARTIFICIAL INTELLIGENCE



# AI UMBRELLA

